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Frequency band determination based on image of communication environment for head-mounted display

Abstract

Methods and apparatus provide for a head-mounted display to be worn by a user located within a physical environment and for engaging in an interactive experience; an entertainment device including a processing section that executes an interactive application that is manipulated in part through receiving inputs from the user; a communication partner operating to relay video and audio signals outputted from the entertainment device to the head-mounted display; continuously acquiring information of the environment at a predetermined frame rate; at least one camera on the head-mounted display, which captures images of the physical environment, where the communication partner is connected to the entertainment device with wired communication, and the communication partner is connected to the head-mounted display with wireless communication.

Inventors: Onishi; Katsuyoshi (Tokyo, JP), Miyazaki; Yoshio (Kanagawa, JP)

Applicant: Sony Interactive Entertainment Inc. (Tokyo, JP)

Family ID: 59090325

Assignee: Sony Interactive Entertainment Inc. (Tokyo, JP)

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Primary Examiner: Talukder; Md K

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of U.S. patent application Ser. No. 17/965,097, accorded a filing date of Oct. 13, 2022, (allowed); which is a continuation of U.S. patent application Ser. No. 17/224,470, accorded a filing date of Apr. 7, 2021, (U.S. Pat. No. 11,528,706, issued on Dec. 13, 2022); which is a continuation application of U.S. patent application Ser. No. 16/903,682, accorded a filing date of Jun. 17, 2020 (U.S. Pat. No. 10,999,847 issued on May 4, 2021); which is a continuation application of U.S. patent application Ser. No. 15/776,242, accorded a filing date of May 15, 2018 (U.S. Pat. No. 10,721,735 issued on Jul. 21, 2020); which is a national stage application of International Application No. PCT/JP2016/087749, filed Dec. 19, 2016; which claims priority to JP Application No. 2015-251360, filed Dec. 24, 2015, the entire disclosures of which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present invention relates to a frequency band determination device, a head-mounted display, a frequency band determination method, and a program.

BACKGROUND ART

(2) There is a known technology for monitoring radio wave conditions for wireless communication and changing a frequency band used for communication in accordance with the monitored radio wave conditions.

SUMMARY

Technical Problem

(3) Millimeter waves, for example, in a 60 GHz band are highly linear. Thus, the communication quality of such millimeter waves significantly varies depending on an environment in which a communication device and a communication partner are placed, namely, for example, depending on whether an obstacle exists in a communication path or the communication device and the communication partner are in the same room. It is therefore preferable that the frequency band used for communication be changed as soon as possible in accordance with a change in the environment in which the communication device and the communication partner are placed.

(4) If, for example, the radio waves used for communication by the communication device are changed to 5 GHz band or other microwaves in a situation where the communication quality degrades while the communication device uses millimeter waves for communication, the communication quality may improve because the microwaves are more diffractive than the millimeter waves. Therefore, if the communication quality is lowered while millimeter waves are used for communication, it is preferable that the radio waves used for communication by the communication device be changed to microwaves as soon as possible.

(5) Consequently, when the communication quality of millimeter-wave communication is monitored and found to be degraded, it is conceivable that an attempt is made to change the radio waves for communication by the communication device to microwaves. However, if such an approach is used, a certain period of time elapses between the instant at which an environmental change occurs to degrade the communication quality and the instant at which the radio waves used for communication by the communication device are changed to microwaves.

(6) The present invention has been made in view of the above circumstances. An object of the present invention is to provide a frequency band determination device, a head-mounted display, a frequency band determination method, and a program that are capable of unprecedentedly quickly changing the frequency band used for communication in accordance with a change in the environment in which the communication device and the communication partner are placed.

Solution to Problem

(7) In order to solve the above problem, a frequency band determination device according to the present invention includes an image acquisition section and a frequency band determination section. The image acquisition section acquires an image of an environment in which at least either a communication device or a communication partner for the communication device is placed. The frequency band determination section determines, based on the acquired image, the frequency band of radio waves used for communication between the communication device and the communication partner.

(8) According to an aspect of the present invention, the frequency band determination section determines the frequency band depending on whether or not the communication device and the communication partner are in the same room.

(9) Alternatively, the frequency band determination section may determine the frequency band depending on whether the acquired image includes an image of the communication device or an image of the communication partner.

(10) Still alternatively, the frequency band determination section determines the frequency band in accordance with the distance between the communication device and the communication partner, which is identifiable from the acquired image.

(11) In this aspect, if the distance is greater than a first threshold value, the frequency band determination section may change the frequency band from a first frequency band to a second frequency band; if the distance is smaller than a second threshold value, the frequency band determination section may change the frequency band from the second frequency band to the first frequency band; the first threshold value may be greater than the second threshold value; the first frequency band may be higher than the second frequency band; and the difference between the second threshold value and the first threshold value may be based on the distance.

(12) A head-mounted display according to the present invention includes an image acquisition section and a frequency band determination section. The image acquisition section acquires an image of an environment in which at least either the head-mounted display or a communication partner for the head-mounted display is placed. The frequency band determination section determines, based on the acquired image, the frequency band of radio waves used for communication between the head-mounted display and the communication partner.

(13) A frequency band determination method according to the present invention includes the steps of: acquiring an image of an environment in which at least either a communication device or a communication partner for the communication device is placed; and determining, based on the acquired image, the frequency band of radio waves used for communication between the communication device and the communication partner.

(14) A program according to the present invention causes a computer to execute procedures for: acquiring an image of an environment in which at least either a communication device or a communication partner for the communication device is placed; and determining, based on the acquired image, the frequency band of radio waves used for communication between the communication device and the communication partner.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating an exemplary overall configuration of a video display system according to an embodiment of the present invention.

(2) FIG. 2A is a diagram illustrating an exemplary configuration of a head-mounted display according to an embodiment of the present invention.

(3) FIG. 2B is a diagram illustrating an exemplary configuration of a relay device according to an embodiment of the present invention.

- (4) FIG. 3A is a schematic diagram illustrating an example in which the video display system according to the present embodiment is disposed in a room.
- (5) FIG. 3B is a schematic diagram illustrating an example in which the video display system according to the present embodiment is disposed in a room.
- (6) FIG. 4 is a functional block diagram illustrating examples of functions implemented in the head-mounted display according to an embodiment of the present invention.
- (7) FIG. 5 is a flowchart illustrating exemplary processing steps performed by the head-mounted display according to an embodiment of the present invention.
- (8) FIG. 6 is a flowchart illustrating exemplary processing steps performed by the head-mounted display according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENT

- (9) An embodiment of the present invention will now be described with reference to the accompanying drawings.
- (10) FIG. 1 is a diagram illustrating an exemplary overall configuration of a video display system **10** according to an embodiment of the present invention. FIG. 2A is a diagram illustrating an exemplary configuration of a head-mounted display (HMD) **12** according to the present embodiment. FIG. 2B is a diagram illustrating an exemplary configuration of a relay device **16** according to the present embodiment.
- (11) As illustrated in FIG. 1, the video display system **10** according to the present invention includes the HMD **12**, an entertainment device **14**, the relay device **16**, a display **18**, a camera/microphone unit **20**, and a controller **22**.
- (12) As illustrated, for example, in FIG. 2A, the HMD **12** according to the present embodiment includes a control section **30**, a storage section **32**, two communication sections **34** (first communication section **34a** and second communication section **34b**), an input/output section **36**, a display section **38**, a sensor section **40**, an audio output section **42**, and a camera section **44**.
- (13) The control section **30** is formed of a microprocessor or other program control device that operates in accordance, for example, with a program installed on the HMD **12**.
- (14) The storage section **32** is formed of a storage element such as a read-only memory (ROM) or a random access memory (RAM). The storage section **32** stores, for example, a program that is to be executed by the control section **30**.
- (15) The first communication section **34a** and the second communication section **34b** are each formed of a wireless local area network (LAN) module or other communication interface having a built-in array antenna. As illustrated in FIG. 1, the present embodiment is configured so that the first communication section **34a** and the second communication section **34b** are disposed on the upper front of the HMD **12**.
- (16) In the present embodiment, the communication standard for communication established by the first communication section **34a** is different from the communication standard for communication established by the second communication section **34b**. For example, the first communication section **34a** establishes 60 GHz band communication, whereas the second communication section **34b** establishes 5 GHz band communication.
- (17) The 60 GHz band communication provides communication at a higher maximum rate than the 5 GHz band communication. Further, the 60 GHz band communication provides higher linearity and a shorter communication range than the 5 GHz band communication. In the present embodiment, either the first communication section **34a** or the second communication section **34b** can be selectively used to communicate with the relay device **16** in accordance with radio wave conditions for wireless communication. Further, a well-known adaptive modulation/coding technology can be used in the present embodiment to adaptively change the modulation method and coding method of the first and second communication sections **34a** and **34b** in accordance with the radio wave conditions for wireless communication.
- (18) The input/output section **36** is formed of an input/output port such as an HDMI (registered

trademark) (High-Definition Multimedia Interface) port, a universal serial bus (USB) port, or an auxiliary (AUX) port.

(19) The display section **38** is formed of a liquid-crystal display, an organic electroluminescence (EL) display, or other display disposed on the front of the HMD **12**, and used to display, for example, video generated by the entertainment device **14**. The display section **38** is accommodated in a housing of the HMD **12**. The display section **38** may receive, for example, a video signal outputted from the entertainment device **14** and relayed by the relay device **16**, and output video represented by the received video signal. The display section **38** according to the present embodiment is capable of displaying a three-dimensional image, for example, by displaying a left-eye image and a right-eye image. The display section **38** may be capable of displaying only a two-dimensional image and incapable of displaying a three-dimensional image.

(20) The sensor section **40** is formed of a sensor such as an acceleration sensor or a motion sensor. The sensor section **40** outputs the results of measurements, such as a rotation amount and movement amount of the HMD **12**, to the control section **30** at a predetermined frame rate.

(21) The audio output section **42** is formed, for example, of a headphone or a speaker, and used to output, for example, a sound that is represented by audio data generated by the entertainment device **14**. The audio output section **42** receives, for example, an audio signal outputted from the entertainment device **14** and relayed by the relay device **16**, and outputs a sound represented by the received audio signal.

(22) The camera section **44** is formed of a digital camera or other camera and used to capture, at a predetermined frame rate, an image of surroundings of a user wearing the HMD **12**. As illustrated in FIG. **1**, the camera section according to the present embodiment includes, for example, four cameras. As depicted, for example, in FIG. **1**, the four cameras are individually disposed on the upper left of the display section **38**, the upper right of the display section **38**, the left side surface of the housing for the HMD **12**, and the right side surface of the housing for the HMD **12**. The cameras may incorporate, for example, a fisheye lens to capture an omnidirectional image of the surroundings of the HMD **12** or may be capable of capturing only a directional image. Further, the camera section **44** may include an infrared camera that is disposed inside the housing for the HMD **12** to detect the line-of-sight direction of the user wearing the HMD **12**.

(23) Furthermore, the camera section **44** may be capable of identifying the distance between the HMD **12** and the relay device **16**. For example, the cameras disposed on the upper left and upper right of the display section **38** may be configured as a stereo camera. Moreover, the distance between the HMD **12** and the relay device **16** may be identifiable based on an image captured by the stereo camera. In addition, the camera section **44** may include an infrared camera or other device (distance measuring device) capable of measuring the distance between the HMD **12** and the relay device **16**, which is disposed forward of the HMD **12**.

(24) The entertainment device **14** according to the present embodiment is formed of a computer such as a game console, a digital versatile disc (DVD) player, or a Blu-ray (registered trademark) player. The entertainment device **14** according to the present embodiment generates video and audio, for example, by executing a stored game program or reproducing content recorded on an optical disk. The entertainment device **14** according to the present embodiment outputs a video signal representative of the generated video and an audio signal representative of the generated audio to the HMD **12** and the display **18** through the relay device **16**.

(25) The relay device **16** according to the present embodiment is formed of a computer that relays the video and audio signals outputted from the entertainment device **14** and outputs them to the HMD **12** and the display **18**.

(26) As illustrated, for example, in FIG. **2B**, the relay device **16** according to the present embodiment includes a control section **50**, a storage section **52**, two communication sections **54** (first communication section **54a** and second communication section **54b**), and an input/output section **56**.

(27) The control section **50** is formed, for example, of a control circuit or a central processing unit (CPU) or other program control device that operates in accordance with a program installed on the relay device **16**.

(28) The storage section **52** is formed, for example, a ROM, a RAM, or other storage element. The storage section **52** stores, for example, a program to be executed by the control section **50**.

(29) The first communication section **54a** and the second communication section **54b** are each formed of a wireless LAN module or other communication interface having a built-in array antenna. As illustrated in FIG. **1**, the present embodiment is configured so that the first communication section **54a** and the second communication section **54b** are included in the front of the relay device **16**.

(30) In the present embodiment, the communication standard for communication established by the first communication section **54a** is different from the communication standard for communication established by the second communication section **54b**. For example, the first communication section **54a** establishes 60 GHz band communication, whereas the second communication section **54b** establishes 5 GHz band communication.

(31) As mentioned earlier, the 60 GHz band communication provides communication at a higher maximum rate than the 5 GHz band communication. Further, the 60 GHz band communication provides higher linearity and a shorter communication range than the 5 GHz band communication. In the present embodiment, either the first communication section **34a** or the second communication section **34b** can be selectively used to communicate with the HMD **12** in accordance with the radio wave conditions for wireless communication.

(32) The input/output section **56** is formed of an input/output port such as an HDMI (registered trademark) (High-Definition Multimedia Interface) port or a USB port.

(33) The display **18** according to the present embodiment is formed, for example, of a liquid-crystal display, and used to display, for example, video represented by a video signal outputted from the entertainment device **14**.

(34) The camera/microphone unit **20** according to the present embodiment includes a camera **20a** and a microphone **20b**. The camera **20a** captures an image, for example, of a subject and outputs the captured image to the entertainment device **14**. The microphone **20b** captures an ambient sound, converts the captured sound to audio data, and outputs the audio data to the entertainment device **14**. Further, the camera **20a** according to the present embodiment is a stereo camera.

(35) The HMD **12** and the relay device **16** are capable of establishing, for example, wireless communication to send and receive data to and from each other. When the HMD **12** and the relay device **16** communicate with each other in the present embodiment, 60 GHz band communication is established between the first communication section **34a** and the first communication section **54a**, and 5 GHz band communication is established between the second communication section **34b** and the second communication section **54b**.

(36) The entertainment device **14** is connected to the relay device **16**, for example, with an HDMI cable or a USB cable. The relay device **16** is connected to the display **18**, for example, with an HDMI cable. The entertainment device **14** is connected to the camera/microphone unit **20**, for example, with an AUX cable.

(37) The controller **22** according to the present embodiment is an operation input device that is used to make an operation input to the entertainment device **14**. The user can use the controller **22** to perform various operation input procedures by pressing a direction key or a button on the controller **22** or tilting an operating stick on the controller **22**. Then, in the present embodiment, the controller **22** outputs to the entertainment device **14** input data associated with the operation input. Further, the controller **22** according to the present embodiment includes a USB port. When connected to the entertainment device **14** with a USB cable, the controller **22** is capable of outputting input data to the entertainment device **14** in a wired manner. The controller **22** according to the present embodiment additionally includes, for example, a wireless communication module,

and is thus capable of wirelessly outputting input data to the entertainment device **14**.

(38) FIGS. **3A** and **3B** are schematic diagrams illustrating examples in which the video display system **10** according to the present embodiment is disposed in a room. In the present embodiment, the first communication section **34a** of the HMD **12** establishes 60 GHz band communication with the first communication section **54a** of the relay device **16** when the HMD **12** and the relay device **16** are in the same room as illustrated in FIG. **3A**. Meanwhile, when the HMD **12** and the relay device **16** are in different rooms as illustrated in FIG. **3B**, the second communication section **34b** of the HMD **12** establishes 5 GHz band communication with the second communication section **54b** of the relay device **16**.

(39) Based on an image that is captured by the camera section **44** to depict an environment in which at least either the HMD **12** or the relay device **16** is placed, the present embodiment determines whether the HMD **12** and the relay device **16** are in the same room or in different rooms. If, for example, an image of the relay device **16** is included in the image captured by the camera section **44**, it is determined that the HMD **12** and the relay device **16** are in the same room. If, by contrast, the image of the relay device **16** is not included in the image captured by the camera section **44**, it is determined that the HMD **12** and the relay device **16** are in different rooms.

(40) If it is determined that the HMD **12** and the relay device **16** are in the same room, the present embodiment exercises control so as to establish 60 GHz band communication between the HMD **12** and the relay device **16**. If, by contrast, it is determined that the HMD **12** and the relay device **16** are in different rooms, the present embodiment exercises control so as to establish 5 GHz band communication between the HMD **12** and the relay device **16**.

(41) As described above, based on the positional relationship between the HMD **12** and the relay device **16**, which is identified from an image captured by the camera section **44**, the present embodiment determines the frequency band of radio waves used for communication.

(42) When the HMD **12** and the relay device **16** are in different rooms, there is a wall, a door, or other obstacle between the HMD **12** and the relay device **16**. Therefore, it is highly probable that 60 GHz or other high-frequency band communication is low in quality. Consequently, when it is determined based on an image captured by the camera section **44** that the HMD **12** and the relay device **16** are in different rooms, the present embodiment exercises control so as to establish 5 GHz or other low-frequency band communication between the HMD **12** and the relay device **16**. Let us assume, for example, that the user wearing the HMD **12** moves from a room in which the relay device **16** is disposed to another room while 60 GHz band communication is established between the HMD **12** and the relay device **16**. In this instance, the communication between the HMD **12** and the relay device **16** switches to 5 GHz band communication. In this manner, the present embodiment changes the frequency band of radio waves used for communication in accordance with a change in the environment in which the HMD **12** and the relay device **16** are placed, and this change occurs earlier than a frequency band change after communication quality confirmation.

(43) Further, millimeter waves are advantageous in that, for example, the maximum rate available for communication is high. Therefore, if the communication quality provided by the millimeter waves is achievable during microwave communication, it is preferable that microwave communication be replaced by millimeter-wave communication. Let us assume here that while 5 GHz band communication is established between the HMD **12** and the relay device **16**, a user wearing the HMD **12** moves from a room to another room in which the relay device **16** is disposed. In this instance, the communication between the HMD **12** and the relay device **16** switches to 60 GHz band communication. In this manner, the present embodiment switches to high-frequency band communication in accordance with an environmental change that improves the quality of high-frequency band communication.

(44) Meanwhile, whether an obstacle exists between the HMD **12** and the relay device **16** to block radio waves may be determined based on an image captured by the camera section **44**. If it is determined that an obstacle exists between the HMD **12** and the relay device **16** to block radio

waves, control may be exercised so as to establish 5 GHz band communication between the HMD **12** and the relay device **16**. If, by contrast, it is determined that no obstacle exists between the HMD **12** and the relay device **16** to block radio waves, control may be exercised so as to establish 60 GHz band communication between the HMD **12** and the relay device **16**.

(45) Further, even if the HMD **12** and the relay device **16** are in the same room, the frequency band of radio waves used for communication may be determined depending on whether an image of the relay device **16** is included in an image captured by the camera section **44**. If, for example, the image of the relay device **16** is included in the image captured by the camera section **44**, control may be exercised so as to establish 60 GHz band communication between the HMD **12** and the relay device **16**. Meanwhile, if, for example, the image of the relay device **16** is not included in the image captured by the camera section **44**, control may be exercised so as to establish 5 GHz band communication between the HMD **12** and the relay device **16**.

(46) Moreover, whether 5 GHz or 60 GHz band communication is to be established between the HMD **12** and the relay device **16** may be determined in accordance, for example, with the distance between the HMD **12** and the relay device **16**. In such an instance, the distance between the HMD **12** and the relay device **16** may be identified by the stereo camera included in the camera section **44**. If the HMD **12** includes a distance measuring device, the distance measuring device may identify the distance between the HMD **12** and the relay device **16**.

(47) Functions of the HMD **12** according to the present embodiment and a process performed by the HMD **12** will be further described below. The HMD **12** according to the present embodiment functions as a frequency band determination device that determines the frequency band of radio waves used for communication between the HMD **12** and the relay device **16** in accordance with an environment in which at least either the HMD **12** or the relay device **16** is placed.

(48) FIG. **4** is a functional block diagram illustrating examples of functions implemented in the HMD **12** according to the present embodiment. The HMD **12** according to the present embodiment need not incorporate all the functions illustrated in FIG. **4**, and may incorporate functions other than those illustrated in FIG. **4**.

(49) As illustrated in FIG. **4**, the HMD **12** according to the present embodiment functionally includes an image acquisition section **60**, a frequency band determination section **62**, and a communication control section **64**. The image acquisition section **60** is mainly implemented by the control section **30** and the camera section **44**. The frequency band determination section **62** is mainly implemented by the control section **30**. The communication control section **64** is mainly implemented by the control section **30**, the first communication section **34a**, and the second communication section **34b**.

(50) The above functions may be implemented by allowing the control section **30** to execute a program that is installed on the HMD **12** formed of a computer and includes commands for the above functions. The program is supplied to the HMD **12**, for example, through an optical disk, a magnetic disk, a magnetic tape, a magneto-optical disk, a flash memory, or other computer-readable information storage medium or through the Internet or the like.

(51) In the present embodiment, the image acquisition section **60** acquires an image of an environment in which, for example, at least either a communication device (the HMD **12** in the current example) or a communication partner for the communication device (the relay device **16** in the current example) is placed. The image acquisition section **60** acquires an image captured by the camera section **44**.

(52) In the present embodiment, based, for example, on an image acquired by the image acquisition section **60**, the frequency band determination section **62** determines the frequency band of radio waves used for communication between the communication device (the HMD **12** in the current example) and the communication partner (the relay device **16** in the current example). Here, as described earlier, depending on whether the HMD **12** and the relay device **16** are in the same room, the frequency band determination section **62** may determine the frequency band of radio waves

used for communication between the communication device the HMD **12** and the relay device **16**. Further, based on the distance between the HMD **12** and the relay device **16**, the frequency band determination section **62** may determine the frequency band of radio waves used for communication between the communication device the HMD **12** and the relay device **16**. Furthermore, depending on whether an image of the HMD **12** or an image of the relay device **16** is included in the image captured by the image acquisition section **60**, the frequency band determination section **62** may determine the frequency band of radio waves used for communication between the communication device the HMD **12** and the relay device **16**. Here, even when the HMD **12** and the relay device **16** are in the same room, the frequency band determination section **62** may determine the frequency band of radio waves used for the communication depending on whether the image of the relay device **16** is included in the image captured by the image acquisition section **60**. Moreover, the frequency band determination section **62** may determine the frequency band of radio waves used for the communication depending on whether the image of the HMD **12** is included, for example, in the image captured by the image acquisition section **60**.

(53) In the present embodiment, the communication control section **64** exercises control so as to establish communication by using the frequency band determined, for example, by the frequency band determination section **62**. If, for example, it is determined that 60 GHz band communication is to be established, the communication control section **64** exercises control so as to let the first communication section **34a** establish communication. If, for example, it is determined that 5 GHz band communication is to be established, the communication control section **64** exercises control so as to let the second communication section **34b** establish communication.

(54) Further, in the present embodiment, the communication control section **64** retains communication section identification data. The communication section identification data is used to manage whether established communication is provided by the first communication section **34a** or the second communication section **34b**. If, for example, the established communication is provided by the first communication section **34a**, the value of the communication section identification data is set to “1.” If, by contrast, the established communication is provided by the second communication section **34b**, the value of the communication section identification data is set to “2.” When the value of the communication section identification data is set to “1,” the communication control section **64** may exercise control so as to stop the second communication section **34b**. Meanwhile, when the value of the communication section identification data is set to “2,” the communication control section **64** may exercise control so as to stop the first communication section **34a**.

(55) The following describes, with reference to the flowchart of FIG. 5, exemplary processing steps that are performed by the HMD **12** according to the present embodiment when the frequency band is to be determined depending on whether the HMD **12** and the relay device **16** are in the same room.

(56) First of all, the image acquisition section **60** acquires an image captured by the camera section **44** (step **S101**). When the camera section **44** is capturing an image at a predetermined frame rate, for example, the latest image captured by the camera section **44** may be acquired in step **S101**.

(57) Then, based on the image acquired in step **S101**, the frequency band determination section **62** determines the frequency band of radio waves used for communication (step **S102**). Here, based, for example, on the image captured by the camera section **44**, the frequency band determination section **62** identifies whether the HMD **12** and the relay device **16** are in the same room. If it is identified that the HMD **12** and the relay device **16** are in the same room, the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 60 GHz band. If, by contrast, it is identified that the HMD **12** and the relay device **16** are in different rooms, the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 5 GHz band. If an image of the relay

device **16** is included in the image captured by the camera section **44**, the frequency band determination section **62** may determine that the HMD **12** and the relay device **16** are in the same room. Meanwhile, if an image of the relay device **16** is not included in the image captured by the camera section **44**, the frequency band determination section **62** may determine that the HMD **12** and the relay device **16** are in different rooms.

(58) Next, the communication control section **64** verifies the frequency band determined in step **S102** and the value of the retained communication section identification data (step **S103**).

(59) If the frequency band determined in step **S102** is the 60 GHz band, and the value of the communication section identification data verified in step **S103** is “2,” the communication control section **64** changes the value of the communication section identification data to “1” (step **S104**). The communication control section **64** then exercises control so as to let the first communication section **34a** establish communication (step **S105**). The process illustrated in the current example then terminates. In step **S105**, the communication control section **64** may start the first communication section **34a** and stop the second communication section **34b**.

(60) If the frequency band determined in step **S102** is the 5 GHz band, and the value of the communication section identification data verified in step **S103** is “1,” the communication control section **64** changes the value of the communication section identification data to “2” (step **S106**). The communication control section **64** then exercises control so as to let the second communication section **34b** establish communication (step **S107**). The process illustrated in the current example then terminates. In step **S107**, the communication control section **64** may start the second communication section **34b** and stop the first communication section **34a**.

(61) In the other cases, the process illustrated in the current example terminates. More specifically, if the frequency band determined in step **S102** is the 60 GHz band, and the value of the communication section identification data verified in step **S103** is “1,” the process illustrated in the current example terminates. Further, if the frequency band determined in step **S102** is the 5 GHz band, and the value of the communication section identification data verified in step **S103** is “2,” the process illustrated in the current example also terminates.

(62) Processing then returns to step **S101**. Subsequently, steps **S101** to **S105** are repeated. Steps **S101** to **S105** may be performed at predetermined time intervals.

(63) Further, as described earlier, in step **S102**, the frequency band determination section **62** may identify, based on the image captured by the camera section **44**, whether an obstacle exists between the HMD **12** and the relay device **16** to block radio waves. If it is identified that an obstacle exists to block radio waves, the frequency band determination section **62** may determine that the frequency band of radio waves used for communication is the 5 GHz band. Meanwhile, if it is identified that no obstacle exists to block radio waves, the frequency band determination section **62** may determine that the frequency band of radio waves used for communication is the 60 GHz band.

(64) Moreover, in step **S102**, the frequency band determination section **62** may determine the frequency band of radio waves used for communication in accordance with the distance between the HMD **12** and the relay device **16**.

(65) The following describes, with reference to the flowchart of FIG. **6**, exemplary processing steps that correspond to the above-described step **S102** and are performed by the frequency band determination section **62** when the frequency band of radio waves used for communication is to be determined based on the distance between the HMD **12** and the relay device **16**.

(66) First of all, based on the image acquired in step **S101**, the frequency band determination section **62** identifies the distance **L** between the HMD **12** and the relay device **16** (step **S201**). Next, the frequency band determination section **62** verifies the value of the communication section identification data retained by the communication control section **64** (step **S202**).

(67) If it is verified in step **S202** that the value of the communication section identification data is “1,” the frequency band determination section **62** verifies whether the distance **L** identified in step

S201 is greater than a first threshold value ($L1+\Delta$) which is obtained by adding the value Δ to a reference distance $L1$ (step **S203**). Here, the reference distance $L1$ is, for example, 8 m, and the value Δ is a value based on the distance L identified in step **S201**, for example, $0.1\times L$. The value Δ may be a value according to image-based recognition accuracy at distance L .

(68) If it is verified in step **S203** that the distance L identified in step **S201** is greater than the first threshold value ($L1+\Delta$) (“Y” at step **S203**), the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 5 GHz band (step **S204**). Subsequently, steps **S103** and beyond are performed. Meanwhile, if it is verified that the distance L identified in step **S201** is not greater than the first threshold value ($L1+\Delta$) (“N” at step **S203**), the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 60 GHz band (step **S205**). Subsequently, steps **S103** and beyond are performed.

(69) If it is verified in step **S202** that the value of the communication section identification data is “2,” the frequency band determination section **62** verifies whether the distance L identified in step **S201** is smaller than a second threshold value ($L1-\Delta$) which is obtained by subtracting the value Δ from the reference distance $L1$ (step **S206**).

(70) If it is verified in step **S206** that the distance L identified in step **S201** is smaller than the second threshold value ($L1-\Delta$), the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 60 GHz band (step **S207**). Subsequently, steps **S103** and beyond are performed. Meanwhile, if it is verified that the distance L identified in step **S201** is not smaller than the second threshold value ($L1-\Delta$), the frequency band determination section **62** determines that the frequency band of radio waves used for communication is the 5 GHz band (step **S208**). Subsequently, steps **S103** and beyond are performed.

(71) In a situation where the above-described scheme is applied, if the distance L is greater than the first threshold value ($L1+\Delta$), the frequency band determination section **62** changes the frequency band of radio waves used for communication between the HMD **12** and the relay device **16** from a first frequency band (e.g., 60 GHz band) to a second frequency band (5 GHz band). Meanwhile, if the distance L is smaller than the second threshold value ($L1-\Delta$), the frequency band determination section **62** changes the frequency band of radio waves used for communication between the HMD **12** and the relay device **16** from the second frequency band (e.g., 5 GHz band) to the first frequency band (60 GHz band).

(72) As described above, the threshold value of the distance between the HMD **12** and the relay device **16** may vary depending on whether the frequency band of radio waves used for communication changes from the first frequency band (high-frequency band) to the second frequency band (low-frequency band) or changes from the low-frequency band to the high-frequency band. Further, the difference ($2\times\Delta$) between the first threshold value ($L1+\Delta$) and the second threshold value ($L1-\Delta$) may be based on the distance L identified in step **S201**.

(73) The present invention is not limited to the above-described embodiment.

(74) For example, all or some of the functions illustrated in FIG. 4 may be implemented by the relay device **16** or the entertainment device **14**. In such an instance, control is exercised to determine whether the relay device **16** uses the first communication section **54a** or the second communication section **54b** to communicate with the HMD **12**.

(75) Further, the frequency band of radio waves used for communication between the HMD **12** and the relay device **16** may be determined based on an image that is captured by the camera **20a** to depict an environment in which at least either the HMD **12** or the relay device **16** is placed. Here, the frequency band of radio waves for communication between the HMD **12** and the relay device **16** may be determined depending, for example, on whether an image of the HMD **12** is included in the image captured by the camera **20a**. If, for example, the image of the HMD **12** is included in the image captured by the camera section **20a**, control may be exercised to establish 60 GHz band

communication between the HMD 12 and the relay device 16. Meanwhile, if, for example, the image of the HMD 12 is not included in the image captured by the camera section 20a, control may be exercised to establish 5 GHz band communication between the HMD 12 and the relay device 16.

(76) Furthermore, the present invention is not only applicable to switching between 60 GHz band communication and 5 GHz band communication, but is generally applicable to switching between high-frequency band communication and low-frequency band communication.

(77) Moreover, the foregoing specific character strings and numerical values and the character strings and numerical values in the accompanying drawings are illustrated and not restrictive.

Claims

1. A head-mounted display comprising: an information acquisition section that continuously acquires information of an environment in which at least one of a communication device and a communication partner for the communication device is placed; and at least one camera which captures images of a physical environment in which a user wearing the head-mounted display is located, wherein: the communication partner receives video and audio signals from an entertainment device and outputs the video and audio signals to the communication device, the information acquisition section continuously acquires the information at a predetermined frame rate, the communication partner is connected to the entertainment device with wired communication, and the communication partner is connected to the communication device with wireless communication using a communication standard selected from a plurality of communication standards based at least in part on images of the physical environment captured by the at least one camera.
2. The head-mounted display of claim 1, wherein the at least one camera includes one or more cameras among a plurality of cameras.
3. The head-mounted display of claim 2, wherein one or more of the plurality of cameras are disposed on the head-mounted display and one or more cameras are disposed in other positions within the physical environment.
4. The head-mounted display of claim 1, wherein the at least one camera captures one of: (i) an omnidirectional image of surroundings of the physical environment, and (ii) a directional image.
5. The head-mounted display of claim 1, wherein the at least one camera operates as a distance measuring device operating to facilitate distance measurements between the head-mounted display and other objects within the physical environment.
6. The head-mounted display of claim 1, wherein the at least one camera includes an infrared camera operating to detect a line-of-sight direction of the user wearing the head-mounted display.
7. The head-mounted display of claim 1, wherein the communication partner is a relay device operating to relay the video and audio signals outputted from the entertainment device to the communication device.
8. The head-mounted display of claim 7, wherein the relay device operates in accordance with a wireless local area network (LAN).
9. The head-mounted display of claim 1, further comprising a frequency band determination section that, upon the acquisition of each information, determines the communication standard from among a plurality of communication standards of frequency bands of radio waves.
10. The head-mounted display of claim 1, wherein the communication device is part of the head-mounted display.
11. The head-mounted display of claim 1, wherein the entertainment device includes a processing section that executes an interactive application that is manipulated in part through receiving inputs from the user of the head-mounted display.
12. A system, comprising: a head-mounted display adapted to be worn by a user for engaging in an

interactive experience; an entertainment device including a processing section that executes an interactive application that is manipulated in part through receiving inputs from the user; and a communication partner operating to relay video and audio signals outputted from the entertainment device to the head-mounted display, wherein the head-mounted display includes: (1) an information acquisition section that continuously acquires information of an environment in which the communication partner is located; and (11) at least one camera which captures images of a physical environment in which the user wearing the head-mounted display is located, wherein: the information acquisition section continuously acquires the information at a predetermined frame rate, the communication partner is connected to the entertainment device with wired communication, and the communication partner is connected to the head-mounted display with wireless communication using a communication standard selected from a plurality of communication standards based at least in part on images of the physical environment captured by the at least one camera.

13. A method, comprising: providing a head-mounted display to be worn by a user located within a physical environment and for engaging in an interactive experience; providing an entertainment device including a processing section that executes an interactive application that is manipulated in part through receiving inputs from the user; providing a communication partner operating to relay video and audio signals outputted from the entertainment device to the head-mounted display; continuously acquiring information of the environment at a predetermined frame rate; and providing at least one camera on the head-mounted display, which captures images of the physical environment, wherein: the communication partner is connected to the entertainment device with wired communication, and the communication partner is connected to the head-mounted display with wireless communication using a communication standard selected from a plurality of communication standards based at least in part on images of the physical environment captured by the at least one camera.

14. At least one non-transitory, computer readable storage medium containing at least one computer program, which when executed by one or more computer processors, causes the one or more processors to perform a method by carrying out actions, comprising: providing a head-mounted display to be worn by a user located within a physical environment and for engaging in an interactive experience; providing an entertainment device including a processing section that executes an interactive application that is manipulated in part through receiving inputs from the user; providing a communication partner operating to relay video and audio signals outputted from the entertainment device to the head-mounted display; continuously acquiring information of the environment at a predetermined frame rate; and providing at least one camera on the head-mounted display, which captures images of the physical environment, wherein: the communication partner is connected to the entertainment device with wired communication, and the communication partner is connected to the head-mounted display with wireless communication using a communication standard selected from a plurality of communication standards based at least in part on images of the physical environment captured by the at least one camera.
